

Digital Twins for Municipal Utilities: Literature Review and Roadmap (OUC Case Study)

Introduction

Municipal utilities like the Orlando Utilities Commission (OUC) are exploring **digital twin** technology to modernize grid operations and improve decision-making. A digital twin is essentially a dynamic **virtual replica** of a physical system that mirrors its real-world counterpart in real time[1]. Unlike static models, a true digital twin remains “alive” – it continuously ingests live data and updates itself as conditions change[2][3]. This means the twin has the “same DNA” as the physical grid and is **synchronized in real-time** with field data, enabling operators to “*fix it before you broke it*” by predicting issues ahead of time[2][3]. Digital twins bridge the physical and digital worlds, allowing utilities to simulate scenarios, forecast outcomes, and optimize performance without disrupting actual operations[4].

Conceptual representation of a power grid digital twin. A digital twin merges real-time operational data with detailed grid models, creating a living virtual model of the network.

In the context of electric utilities, digital twins promise several benefits. They provide a **real-time, holistic view** of infrastructure, from generation units down to customer meters, enabling proactive maintenance and operational efficiency[5]. For example, operators can use a generator’s twin to simulate performance under various loads and identify signs of wear or inefficiency, scheduling maintenance before a failure occurs[4]. Likewise, a distribution grid twin can predict overloads or voltage issues in advance, helping prevent outages and improve reliability[5]. Underpinning these capabilities are advanced analytics and machine learning: digital twins leverage **AI/ML** to analyze sensor data, learn normal vs. anomalous patterns, and continuously improve their predictive accuracy[6]. In short, digital twin technology is emerging as a cornerstone of utility innovation, offering OUC a path to optimize asset management, minimize downtime, and enhance customer service through data-driven insights[7].

Core Concepts of Digital Twins in Electric Utilities

Digital twins in the electric utility domain are defined by high fidelity and integration. At their core, they are “**faithful, testable mirror[s] of the grid**” – precise virtual models that can be probed and stressed in ways the real grid cannot[8]. This fidelity comes from blending **physics-based models** of electrical equipment with streaming field data. In practice, a power grid twin incorporates detailed electrical models (generators, transformers, lines, protection devices, etc.) and aligns them with real measurements from SCADA systems, sensors, and historical data archives[9]. The twin is continually calibrated so that its simulated state matches the actual grid within tight error bounds[10].

In effect, the digital twin behaves like a **real-time simulator** of the grid that is kept accurate by live data feeds.

Crucially, a utility's digital twin is not a one-time static snapshot but a living model that evolves with the system. Industry experts emphasize that a twin must be *integrated but separate* – it runs on its own platform (often cloud-based), yet is fed by and reflective of the live system[11][12]. The twin updates automatically whenever the real network's state changes, ensuring analyses are based on the current reality[2][3]. This dynamic coupling allows the twin to serve as a **real-time decision support tool**. Operators and planners can ask “*what-if*” questions of the twin – for instance, “*What happens if this substation transformer trips?*” or “*How would high EV charging adoption impact our grid?*” – and the twin will respond based on up-to-date conditions. Because the twin runs on high-performance computing infrastructure, it can simulate complex interactions at engineering timescales faster than real time[8]. This provides clarity into grid behavior that would be impossible to test on the live system, shortening planning cycles and reducing risk by revealing issues before they occur[8][13].

From a data science perspective, these core concepts mean that an electric utility's digital twin combines **domain physics and data analytics**. It is a convergence of electrical engineering models with big data from IoT sensors. Machine learning algorithms are often embedded to calibrate the model, detect anomalies, and predict trends[6]. The result is a powerful hybrid of first-principles simulation and data-driven insight – effectively an AI-enhanced simulator that grows more accurate over time. For OUC's data scientists, understanding both the physical grid behavior and the data patterns is essential to fully leverage the twin for predictive analytics and operational optimization.

High-Fidelity Grid Modeling and Physical Grid Simulation

One defining feature of a digital twin is its **grid fidelity** – the level of detail and accuracy with which it represents the physical grid. High fidelity modeling means capturing the electrical characteristics of each component (generators, inverters, lines, relays, etc.) and the network topology with great precision. The literature stresses that a digital twin in power systems is a “*high fidelity virtual counterpart linked to live data sources and engineering workflows*”[9]. It **blends physics-based models** of equipment with real-world measurements (e.g. SCADA telemetry, phasor measurement unit data, historical load records) to ensure the virtual grid behaves like the real one[10]. Through continuous calibration against field sensor readings, the twin can track the actual grid's state within tight error margins[14]. This gives utility engineers confidence that simulations on the twin will closely mirror outcomes on the physical grid.

Achieving such fidelity involves sophisticated **physical grid modeling**. The twin typically runs detailed power flow and dynamic simulations. It can model voltage, current, power flows, and even electromagnetic transients if needed, across the network. Protective device logic (relay settings, breaker operations) and control systems are also emulated to see how the grid responds to faults or switching events. For example, OUC's digital twin

could include a full model of a substation – down to transformers, capacitor banks, and relays – allowing precise simulation of fault scenarios or load switching at that substation. High-performance computing is used so that these complex simulations can run in (or faster than) real time. In fact, the twin can be connected to real control hardware in a **Hardware-in-the-Loop (HIL)** setup to test equipment under simulated grid conditions[15]. This is extremely valuable for verifying new control algorithms or DER integrations in a safe virtual environment before field deployment.

Moreover, high-fidelity twins support **multi-scale modeling** of the grid. The scope can range from a single feeder or microgrid up to the entire transmission system[15]. Engineers can zoom into a specific component or look at system-wide behavior. By matching the physics, timing, and data quality of the real grid, the twin reveals how small changes (like adjusting a voltage setpoint) ripple through the system[16]. Planners can study *N-1* or *N-2* contingencies (simulating one or two elements out of service) on the twin to assess reliability, which is far safer and more thorough than relying on past experience alone[17]. In short, **grid fidelity modeling** means the digital twin is as detailed as necessary to reflect reality, enabling true “virtual testing” of grid operations and reinforcement plans. For a data scientist, this high-fidelity simulation environment provides a rich sandbox to run experiments, generate synthetic data for training algorithms, and validate ML model predictions against physical laws.

Data Sources and Sensors for a Digital Twin

Implementing a digital twin requires tapping into a wide array of **data sources and sensors** to feed the model. High-fidelity modeling is only possible if the twin has accurate, real-time and historical data about the grid’s state. Key data inputs and sensing infrastructure include:

- **SCADA Telemetry:** Supervisory Control and Data Acquisition systems provide real-time measurements from substations and feeders (voltages, currents, power flows, breaker statuses, etc.). SCADA acts as the nervous system of the grid, streaming operational data that the twin uses to mirror current conditions[10]. For example, OUC’s SCADA might report a feeder’s load at 4-second intervals; the twin ingests this to update the virtual model’s load on that feeder accordingly.
- **Intelligent Electronic Devices & PMUs:** Modern substations have intelligent devices and **phasor measurement units (PMUs)** that sample electrical waveforms at high speed (30+ samples per second). These provide time-synchronized voltage and current phasors across the grid, enabling the twin to capture dynamic phenomena (like oscillations or transient events) with high granularity[10]. PMU data enhances the twin’s fidelity for stability analysis and model calibration.
- **Advanced Metering Infrastructure (AMI):** Smart meters at customer premises (AMI) deliver granular consumption data (hourly or 15-min interval usage per customer). This data is crucial for modeling **customer load behavior** on the

distribution network. By integrating AMI data, the digital twin can simulate load patterns on each feeder or even each transformer, improving accuracy of load forecasts and voltage profiles. In fact, advanced use cases involve creating a digital twin of each customer's demand profile for targeted analysis[18][19].

- **Geographic Information System (GIS) Data:** GIS provides the **network model and asset registry** – the connective tissue for the twin. It contains the electrical connectivity (which customer is on which transformer, how feeders are routed), equipment attributes (sizes, ratings, installation dates), and geospatial location of assets. GIS is foundational for any digital twin because it gives structure and context to all other data[20][21]. Essentially, GIS data allows the twin to be a true **spatial replica** of the network, so that sensor readings and simulations can be tied to specific components and locations on the grid.
- **Operational Systems and Sensors:** A variety of other operational technology (OT) systems contribute data. **Outage management systems (OMS)** provide information on outages and switch states, useful for the twin to reflect network reconfigurations. **Distributed energy resource (DER) monitors** (like solar inverter telemetry or battery management systems) feed the twin with real-time generation from rooftop PV, EV chargers, or storage devices. **Line sensors and IoT devices** on the distribution lines (reporting metrics like conductor temperature, sag, or fault currents) give the twin fine-grained visibility into conditions along the feeders. **Substation sensors** (for transformer temperature, SF6 breaker gas pressure, etc.) support asset health modeling in the twin.
- **Historical and Enterprise Data:** To complement real-time feeds, the twin ingests historical data archives (e.g. past load curves, weather data, maintenance logs). These data are used for training predictive models and scenario analysis. Enterprise IT systems like **asset management databases** (which track equipment age, maintenance history) and **customer information systems** (for customer classes, critical customers, etc.) also integrate into the twin's data ecosystem. The result is a **comprehensive data picture**: one recent guide notes that digital twins are effectively an **integration of IT, OT, and engineering technology**, bringing together previously siloed datasets into one unified model[22].

Building robust data pipelines and ensuring data quality is a major part of twin implementation. A data scientist at OUC would likely spend significant effort on **data cleaning, validation, and fusion** across these sources to create a reliable single source of truth. Indeed, experts highlight that access to “*vast amounts of cleaned and organized data*” is a core requirement for a functioning digital twin[18]. Without trustworthy data from sensors and systems, the twin's fidelity and usefulness would quickly degrade. Therefore, establishing streams from SCADA, AMI, GIS, and more into a common data platform (often a cloud-based data lake or time-series database) is one of the first steps in standing up the digital twin.

Integrating IT/OT Systems (SCADA, GIS, AMI, Enterprise Platforms)

Implementing a digital twin at a utility like OUC is as much an **integration project** as it is a modeling project. It requires bridging the traditional divide between **operational technology (OT)** – systems that run the grid in real time – and **information technology (IT)** – systems that manage business and planning data. The twin will sit at the intersection of these, consuming data from both and providing insights to both. Successful integration means that the twin becomes a “*shared source of truth*” that all departments (planning, operations, maintenance, customer service, etc.) can utilize[23].

Key integration points include:

- **SCADA/ADMS Integration:** The twin must interface with SCADA (and/or Advanced Distribution Management Systems) to pull in real-time telemetries and also to reflect switching operations. For example, if operators open a switch or reconfigure feeders, the ADMS would inform the twin so that its network model updates. This can be done via APIs or historian databases that store SCADA data. A typical digital twin dashboard will integrate with the SCADA historian for a live feed of analog measurements[24].
- **GIS and Network Model Integration:** The utility’s GIS or network modeling software (which might be part of an ADMS or separate) provides the baseline electrical model for the twin. Integration here ensures that any changes in the physical network (new lines, re-phasing, asset replacements) are propagated to the twin’s model. Many utilities treat their GIS connectivity model as the system of record; the twin can subscribe to updates or periodically sync with GIS data so that it always mirrors the “as-built” network.
- **AMI and Meter Data Management (MDM):** Connecting the twin to the meter data system allows it to incorporate detailed load and voltage data at customer endpoints. For instance, OUC’s twin might integrate with its MDM to pull daily load profiles for each feeder or transformer. This enhances planning studies and load forecasting within the twin, and also allows *near-real-time* anomaly detection (e.g. voltage fluctuations at the end of a line as reported by smart meters).
- **Outage Management and DER Systems:** The twin should also connect with the Outage Management System (OMS) to know about outages, crew actions, and restoration status. OMS integration means the twin can simulate current outage conditions or predict network sections at risk. Similarly, integration with DER management platforms (like a **DERMS**) provides real-time visibility and control statuses of distributed resources, which the twin needs for accurate simulations of scenarios like high solar output or demand response events[23].
- **Enterprise IT Platforms:** On the IT side, the twin will draw from databases like asset management (for equipment health and maintenance records), **GIS (spatial context)**, and even weather or financial systems for advanced use cases.

Integration could involve ETL processes to periodically import data (e.g., asset failure rates, work orders) into the twin's analytical platform. The goal is to break down silos: one industry roadmap describes integrating outage, AMI, ADMS, SCADA, and other systems into a unified dashboard for the digital twin[24]. By doing so, the twin aggregates all **key utility data streams** in one place.

Achieving IT/OT integration comes with challenges in data formats, protocols, and departmental coordination. Standards like the **CIM (Common Information Model)** can help align data models across systems. Middleware or enterprise service buses might be used to interface real-time OT data with cloud analytics environments. Cybersecurity (discussed later) is also critical when bridging OT and IT networks. For a data scientist, a well-integrated twin opens up access to rich datasets that were historically separate – meaning better training data for models and the ability to build features that combine, say, SCADA sensor trends with maintenance history and weather, all in one analytical environment. In summary, integration ensures the digital twin isn't an isolated tool, but rather an *enterprise-wide asset* that ties together SCADA, GIS, AMI, and more into a comprehensive, spatially-aware view of the utility's operations[21][22].

Roadmap for Implementing a Digital Twin at OUC

Implementing a digital twin is a complex, multi-year journey. It's best approached in **phases**, starting small and building in capability and scope over time[25]. Below is a phased roadmap tailored for OUC, covering development stages from initial pilot to full deployment. At each stage, considerations around data ingestion, real-time synchronization, analytics, and governance are highlighted.

Phase 1: Pilot Project and Foundations

Scope: Begin with a narrowly focused pilot to demonstrate the digital twin concept and build internal expertise. Many utilities start by developing a twin for a single substation, a handful of feeders, or one critical asset class[26]. OUC might choose a pilot like a “**digital substation**” twin for a high-load substation or a **power plant twin** for one generation unit at the Stanton Energy Center.

Objectives: In this phase, the goal is to implement the core twin platform and achieve basic functionality: - Stand up the **data infrastructure** needed for the twin. This includes setting up data pipelines from key sources (SCADA historian, GIS extracts, meter data). Initial data ingestion might be batched or semi-manual at first. The team will work on data cleansing and alignment (e.g., matching SCADA points to model components) to populate the twin's database. - Develop the **baseline simulation model** for the pilot scope. For a substation twin, this means modeling the one-line diagram, connected feeders, and nearby DERs in a simulation tool or platform. Validate the model against known power flow or short-circuit calculations. - Implement **real-time synchronization** on a small scale. For example, have the twin pull SCADA readings from that substation every few minutes to update the model's state (voltage levels, breaker statuses). The pilot doesn't need ultra-

low latency; even near-real-time (say, every 5 minutes) updates can show the concept. - Build a simple **user interface or dashboard**. This could be a web-based dashboard showing the substation schematic with live values, or an analytic workspace where engineers can run what-if studies on the pilot model. - **Analytics Layer (pilot-level)**: Introduce one or two analytical applications to prove value. For instance, implement an **overload prediction**: using the twin plus historical load data, predict if any feeder will exceed capacity under a heat wave scenario. Or a **what-if scenario**: if a particular feeder is opened, how are loads transferred? Even a basic **voltage drop analysis** using the twin could demonstrate improved insight.

During this phase, the organization also identifies gaps in data or skills. It serves as a *learning period* to understand how the twin integrates with existing systems and what data quality issues might arise. Early wins and visualizations from the pilot will help secure buy-in from stakeholders. Crucially, a pilot can be executed with relatively low risk and investment, but provides a blueprint for next steps[27]. Documentation from this phase (the roadmap, data inventory, initial architecture) will guide the scaled rollout[28][24].

Phase 2: Expanded Deployment and Integration

Scope: In this phase, the digital twin expands to cover a **broader portion of the grid** and brings in more data streams. For OUC, this might mean extending the twin from one substation to an **entire district or feeder group**, or from one power plant unit to an **entire generation fleet**. Essentially, the twin evolves from a niche pilot to a more representative cross-section of the utility's operations.

Objectives: Key goals and activities in Phase 2 include: - **Scaling Data Ingestion:** Move from pilot-scale data feeds to enterprise-scale. This could involve deploying a streaming data platform (e.g. Kafka or an IoT hub) to handle real-time data ingestion from hundreds of feeders and thousands of devices. The twin's data architecture might be hardened for high throughput, ensuring it can ingest AMI data from all smart meters, SCADA from all substations, etc., in real or near real time. Data retention and management strategies (data lake, time-series database) are implemented to store historical data for analytics. - **Full Integration of IT/OT Systems:** Integrate additional systems that were out-of-scope in the pilot. For example, bring in Outage Management System data so the twin knows about outages and switching in the field[24]. Connect the enterprise asset management system so that asset condition indices or maintenance schedules are linked to the twin's components. Ensure GIS and network model updates flow automatically into the twin as the asset base changes. At this stage, the twin should become a **convergence point** for all relevant operational data streams in the utility. - **Real-Time Synchronization:** Increase the frequency and fidelity of sync with live data. By now, the twin should update continuously or at least at SCADA scan rates for key parameters. If Phase 1 was updating every few minutes, Phase 2 might target sub-minute updates or event-driven updates (e.g., whenever a breaker status changes, the twin reflects it immediately). The twin essentially runs in parallel with the live grid, ingesting a firehose of telemetry. This might necessitate moving the twin's computation to a scalable cloud environment for elasticity[29]. - **Enhancing the**

Simulation Model: Expand the model to the new scope, adding more feeders, more detailed generator models, etc. Also improve fidelity – e.g., incorporate dynamic models for frequency and voltage control, add weather correlation to load models, include a larger portion of the protection system. Perform rigorous validation: the twin’s power flow results should be cross-checked against EMS or planning studies to ensure accuracy as complexity grows. - **Analytics Layer – Advanced Applications:** Layer more advanced analytics and AI use cases on the now larger twin. For instance, implement a **load forecasting model** that uses the twin’s data (AMI, weather, historical loads) to predict peak loads for the next day or season. Or a **DER impact analysis** tool that uses the twin to simulate high solar scenarios and uses machine learning to identify optimal locations for new DERs. Another high-value use case is **predictive maintenance**: use sensor data in the twin to train ML models that predict transformer or feeder failure risk, effectively creating a risk heat map of the grid. These analytics can run continuously on the twin’s latest data – for example, an outage prediction algorithm that constantly evaluates the twin’s state against patterns preceding past failures. - **User Adoption and Training:** With a broader twin, more teams at OUC will interact with it. Phase 2 should involve training operations staff, planners, and data analysts on how to use the twin’s dashboards and tools. Their feedback will be critical to refine functionality. For instance, control room operators might want a simplified real-time view with alarms for anomalies, whereas planning engineers might request a mode to simulate long-term scenarios. Governance processes may start to form around how model changes are managed and how users request new features.

By the end of Phase 2, the digital twin should demonstrate tangible benefits in multiple domains (e.g., fewer outages from better switching decisions, or cost savings from deferred capital upgrades as seen in scenario simulations[30]). It essentially becomes an **operational analytics platform** that stakeholders trust for insights. This paves the way to institutionalize the twin as a permanent part of the utility’s toolkit.

Phase 3: Full-Scale Deployment and Optimization

Scope: In Phase 3, the digital twin reaches **enterprise scale** – encompassing all major elements of OUC’s electric system (and potentially water system, if desired) and fully embedded in the utility’s processes. The twin moves from a project status to a sustained platform for operations, planning, and enterprise analytics.

Objectives: Key aspects of this mature stage include: - **Complete Coverage:** The twin now models **generation, transmission, and distribution** assets of OUC with high fidelity. All substations, feeders, and critical assets are represented. The model is kept current with every network change. This may involve automated data pipelines from engineering design software whenever upgrades or expansions occur. - **Real-Time Operation & Control Integration:** The twin could be integrated into control room operations. For example, operators might have a **digital twin console** next to the SCADA/ADMS console, using it for contingency analysis on the fly. Real-time synchronization is fully achieved; the twin updates state as fast as data comes in (possibly seconds or sub-second for PMU streams). In advanced setups, the twin might even send control suggestions to operators (e.g., “if

voltage at Bus X remains low, consider switching capacitor bank Y” based on twin simulation). The twin essentially functions as the real-time “shadow” of the grid. - **Enterprise Analytics Hub:** All analytics applications are layered on the twin’s data. This includes real-time dashboards for asset health, predictive algorithms for outages and load, and even customer analytics. For instance, the marketing or demand-side management team could use the twin to target energy efficiency programs by querying “*which neighborhoods have the highest peak loads and could benefit from demand response?*”, leveraging the twin’s detailed load models[31][32]. The twin becomes a central platform where AI/ML models are deployed and run continuously, providing decision support across departments. - **Governance and Maintenance:** At full deployment, a formal **governance structure** oversees the digital twin. This includes regular model validation checks, data quality monitoring, and procedures for updating model parameters. A dedicated team (potentially including data scientists, power engineers, and IT specialists) administers the twin, ensures it’s cyber-secure, and interfaces with all business units for new use cases. The governance would also handle the twin’s lifecycle – keeping software up to date, scaling infrastructure as data grows, and ensuring compliance with any regulations (like NERC CIP cybersecurity standards for any connection to the OT network). - **Continuous Improvement:** Even at scale, the twin should continuously evolve. Phase 3 would institute a feedback loop where lessons from operations feed into model improvements. For example, if an outage occurred that was not predicted by the twin, an investigation is done and the model or algorithm is adjusted to capture that scenario next time. The utility may also incorporate new technologies as they arise: for instance, if new **sensors or IoT devices** get deployed (perhaps pole-mounted fault detectors or advanced EV charging data), those are integrated. The twin can also extend beyond the grid – e.g., linking with a **city digital twin** for holistic city-utility simulations (as Orlando city itself is exploring digital twins for infrastructure planning).

By the end of Phase 3, OUC’s digital twin would be a **fully realized digital platform** underpinning a smarter utility. All data is ingested and synchronized in real time, analytics are layered throughout, and the twin informs both day-to-day operations and long-term planning strategies. The journey from pilot to full deployment might take several years, but it can be expedited by focusing on high-value use cases at each phase to demonstrate ROI and by iteratively scaling up. Notably, a successful pilot can serve as the blueprint – “*a roadmap for broader implementation*” – allowing OUC to scale thoughtfully based on demonstrated results and organizational readiness[27].

Data Architecture, Cybersecurity, and Governance Considerations

Throughout the above phases, certain cross-cutting elements must be addressed to ensure the digital twin’s success and sustainability. Three critical areas are **data architecture, cybersecurity, and governance**:

- **Data Architecture:** A robust data architecture is the backbone of the digital twin. This includes the hardware/cloud infrastructure and software stack that collect, store, and serve data to and from the twin. Early on, OUC should decide on a

platform (on-premises vs. cloud or hybrid) for hosting the twin's simulations and data. Cloud infrastructure is attractive for its scalability – for example, running millions of simulations or hosting millions of “mini-twins” (one for each customer or device) is only feasible with elastic cloud computing[29]. However, latency and on-prem control are considerations if the twin is to be tightly integrated with real-time operations. A common approach is a hybrid architecture: core real-time data pipelines and control remain on-prem, while heavy analytics and historical data reside in a cloud-based data lake or enterprise analytics platform. The data architecture should support **real-time streaming** (for live sensor feeds) as well as **batch processing** (for large-scale training of ML models on historical data). Using open data standards and a unified schema (e.g., CIM for grid model, time-series standards for telemetry) will future-proof the system and ease integration. Since the twin will accumulate enormous volumes of data (SCADA streams, AMI readings, simulation outputs), big data management techniques are needed – efficient querying, archiving older data, and perhaps edge processing to reduce raw data flows. AI can even assist here: utilities are looking at AI to **automate data cleanup and synchronization for higher-fidelity twins**[33].

- **Cybersecurity:** Given that a utility digital twin interfaces with operational data and possibly controls, it is a **critical asset** that must be secured. A breach or tampering of the twin could not only expose sensitive grid data but, in worst cases, could mislead operators or be used to predict and exploit grid vulnerabilities. Therefore, strong cybersecurity measures are non-negotiable. This includes **network security** (ensuring data coming from OT systems to the twin travels over secure, possibly one-way, channels to isolate critical control systems), **access control** (only authorized personnel and applications can access the twin's data and functions, with role-based permissions), and **monitoring** (continuous monitoring for any anomalies or intrusions in the twin environment). Some research even proposes using **blockchain** or distributed ledger technology to enhance the security and trustworthiness of digital twin data exchanges[34]. At a minimum, OUC's twin should comply with industry cybersecurity standards and best practices, such as encryption of data in transit and at rest, secure API endpoints, and regular security audits. Additionally, if the twin enables any automatic control actions (e.g., a closed-loop self-healing function), there must be safeguards to prevent false triggers or malicious commands from causing grid disturbances. Cyber resilience drills (simulating cyberattacks on the twin to test its isolation from operational control) might be part of governance in later phases.
- **Data Governance and Model Governance:** As the twin becomes an enterprise asset, formal governance is needed to manage its evolution. **Data governance** ensures that the data feeding the twin remains high-quality, consistent, and compliant with privacy regulations. For instance, policies should define how AMI (customer usage) data is used and shared in analyses to protect customer privacy. Data retention policies will dictate how long data is kept and when it's purged or

anonymized. A governance committee might oversee integration of any new data source, verifying its accuracy and relevance. On the **model governance** side, procedures are needed for updating the digital twin's models and algorithms. If a protection setting in the field changes or a new feeder is added, how and when is the twin updated? Version control for the model configuration is important – the twin's fidelity relies on it being in sync with the latest system configurations. Equally important, any AI/ML models running on the twin (for forecasting, asset failure prediction, etc.) need governance: they should be periodically retrained with new data, validated for bias or errors, and recalibrated if the physical world changes (e.g., a new type of DER proliferates making old load patterns obsolete). An interdisciplinary governance team (data scientists, grid engineers, IT, and cybersecurity staff) can meet regularly to handle these changes in a controlled manner. This ensures the digital twin remains **accurate, secure, and aligned with business goals** over time.

Establishing strong architecture and governance from the start is what turns the digital twin from a flashy demo into a reliable, long-term utility asset. Utilities that have embarked on this journey note that maximizing the value of digital investments requires integration and **data sharing across teams, while maintaining cybersecurity**^[35] – essentially echoing that technology alone isn't enough; process and policy must evolve in tandem. For a data scientist at OUC, participating in these governance efforts is key: it provides oversight on data quality (critical for model performance) and a say in prioritizing which analytical features the twin should develop next. Involvement in governance also ensures the data science perspective (e.g., the need for certain data or sandbox environments for experimentation) is considered in strategic decisions.

AI and Machine Learning Use Cases for Utility Digital Twins

A fully developed digital twin opens the door to numerous **AI and machine learning (ML)** applications in generation, transmission & distribution (T&D), distributed energy resources (DERs), and customer analytics. These use cases are particularly exciting for data scientists, as they involve applying advanced algorithms to solve real-world utility problems using the twin's rich data. Below, we outline several high-impact AI/ML use cases relevant to OUC's generation fleet, grid network, and customers:

- **Generation Asset Optimization and Predictive Maintenance:** Digital twins of power generation assets (like turbines, boilers, solar farms) allow for ML-driven predictive maintenance and efficiency optimization. By continuously monitoring sensor data (vibrations, temperatures, pressures) through the twin, ML models can detect early signs of equipment stress or degradation. For example, a twin of a gas turbine can use weather and operational data to predict parts wear and schedule maintenance before a failure occurs^[4]. AI algorithms can also optimize generation dispatch by simulating different operating conditions in the twin to minimize fuel use or emissions. In practice, such asset-focused digital twins have shown significant value – **AI-driven twins have helped monitor thousands of critical**

assets and reportedly saved over \$1.6 billion by preventing failures in the energy sector[36]. For a data scientist, this use case involves time-series analysis, anomaly detection, and remaining useful life modeling on equipment data.

- **T&D Grid Reliability and Self-Healing:** On the transmission and distribution side, AI can leverage the twin to improve reliability and enable self-healing grid actions. One application is **fault prediction** – using historical outage data, weather feeds, and real-time sensor readings (like momentary fault indicators), ML models can predict which lines or transformers are likely to fail during a looming storm or under stress. The twin can simulate the impacts and identify the optimal grid reconfiguration to minimize customer impact. With sufficient accuracy, such a system could pre-emptively sectionalize or adjust protection settings to isolate likely fault areas, embodying a self-healing network. Research indicates that digital twins combined with AI could detect anomalies like **overheated lines or abnormal losses** and take decision-making a step further into automated grid reconfiguration[37]. In essence, the twin + AI would act as a real-time reliability coach for the grid. Data science work here includes classification models for fault likelihood, clustering of sensor anomaly patterns, and possibly reinforcement learning for deciding optimal switching actions.
- **Distributed Energy Resource (DER) Integration and Optimization:** As DERs (solar PV, batteries, EV chargers) proliferate, utilities face challenges in forecasting their impact and orchestrating them for grid stability. A digital twin augmented with AI is a powerful tool for DER integration. For instance, AI models can **forecast solar generation** and EV charging demand in different parts of the network using weather data and usage patterns, feeding this into the twin to simulate tomorrow's grid conditions. Planners can evaluate scenarios of high DER penetration on the twin, and use optimization algorithms (like genetic algorithms or reinforcement learning) to find the best placements for energy storage or new feeders. One report noted that initially, digital twins in energy were used for *“basic supply planning around DERs”*, but now utilities realize these twins can be used from high-level transmission planning down to distribution and customer-level DER management[38][32]. An example use case: identifying feeders at risk of backfeeding due to rooftop solar and using the twin to virtually test mitigation measures (capacitors, new regulators) before physical deployment. Data scientists would develop forecasting models, scenario analysis tools, and possibly real-time control algorithms for DER dispatch (e.g., using AI to decide how much battery storage to charge or discharge based on twin simulations of grid conditions).
- **Customer Load Forecasting and Demand Management:** A particularly data-rich use of digital twins is in **customer energy analytics**. By leveraging AMI data and detailed building models, a utility can create digital twins of customer loads (or even each premise). AI can then be applied to forecast loads at various levels – individual customer, feeder, or system peak – with higher accuracy. The twin provides context like the network constraints, so forecasts can be adjusted not just

for weather and behavior, but also for grid topology (e.g., voltage drop might temper demand in certain areas). A real-world example saw a utility modeling **2.3 million digital twins – one per customer – to forecast technology adoption and demand over 30 years**, using cloud computing to run these massive simulations[29]. With such models, utilities can query the twin with questions like *“What are the top 10 locations in our territory with the highest peak demand, and what demand-response or efficiency programs could reduce that?”*[39][40]. The answers inform targeted program design, such as focusing incentives on specific neighborhoods. For a data scientist, this use case involves big data handling, building physics models (for buildings’ response to weather), and applying AI techniques from time-series forecasting to agent-based modeling (simulating how customers adopt technologies like EVs or solar under different incentives).

- Outage Prediction and Rapid Restoration:** Beyond predicting asset failures, AI can help anticipate and manage broader outages. By integrating weather forecasts (e.g., hurricane paths, lightning probability), vegetation data (tree trim records, satellite imagery), and real-time grid stress indicators, ML models can give an **“outage risk score”** for different grid segments ahead of time. The digital twin can then be used to simulate the effect of potential outages (which areas go dark if a certain line falls) and evaluate contingency plans. This guides pre-staging of crews and equipment to areas of highest risk, improving resilience. During an actual outage, the twin can assist in **fault location** (comparing telemetered values to isolate the fault section) and **restoration optimization** (testing switching sequences to restore power fastest without overloading other parts). AI techniques like decision-tree models and Bayesian networks have been used in some utilities for outage predictions. The twin provides the canvas to run these predictive models and then play out “what-if” restoration paths in a risk-free manner.
- Asset Health Monitoring and Investment Planning:** Aggregating all the condition data in the twin allows for ML-driven **asset health indices** and long-term planning insights. For example, by using temperature, loading, and oil gas sensor data from transformers in an ML model, the twin can continuously score each transformer’s health and predict its remaining life. Utilities like GE have implemented such AI-based digital twin models to catch subtle anomalies in asset behavior, reducing unplanned downtime significantly[36]. On a planning horizon, the twin can simulate asset aging and failure probabilities under various maintenance scenarios. AI optimization can then help schedule maintenance or replacements for *just-in-time* reliability, maximizing asset life and deferring unnecessary capital spend. A cited benefit is more informed investment planning – *utilities can simulate scenarios like equipment degradation to decide where and when to invest in upgrades, deferring unnecessary capex*[41]. For data scientists, developing predictive maintenance models is a key task (using techniques like survival analysis or neural networks on time-series sensor data), as is working with planners to run multi-year simulations

on the twin and extracting insights (using statistical analysis or AI optimization to recommend investments).

Each of these use cases illustrates how the digital twin serves as a **platform for AI/ML** in the utility. The twin provides the integrated data and physics context, and AI provides the advanced pattern recognition and predictive muscle. A data scientist at OUC would find themselves at the center of this, responsible for developing and deploying these models. This involves typical data science work (data wrangling, feature engineering, model training) but also a deep collaboration with engineers to ensure the models respect physical realities and produce actionable results. The payoff is significant: from reducing outages and maintenance costs to engaging customers with personalized insights, AI-enabled digital twins can drive utility performance to new heights[42][43]. As one industry expert put it, *“we now have the computational power to make digital twins of everyone and everything”*[44] – the challenge and opportunity for data scientists is to harness that power on meaningful utility problems, ultimately delivering a smarter, more resilient grid for OUC and its customers.

Conclusion

In summary, implementing a digital twin at OUC involves combining cutting-edge technology with careful planning and cross-departmental strategy. The literature review highlights that digital twins are **dynamic, high-fidelity virtual models** of utility systems that, when fed by real-time data, become powerful tools for simulation, prediction, and optimization[45][9]. Core components include detailed physical grid models, integration of myriad data sources (SCADA, GIS, AML, etc.), and advanced analytics layers that turn data into actionable intelligence. A phased roadmap – starting from a focused pilot and scaling up in stages – is the prudent way to implement such a complex system, allowing the utility to realize early benefits, learn, and build out the twin’s capabilities in alignment with organizational readiness[26][27]. Key considerations around data architecture, cybersecurity, and governance must be embedded in each phase to ensure the twin’s reliability and security as it becomes an integral part of operations.

For a data scientist at OUC, the digital twin initiative is an exciting frontier. It transforms the utility into a **data-rich environment** where physics-based simulation and AI converge. Data professionals will play a pivotal role in integrating data sources, developing machine learning models for uses from load forecasting to asset failure prediction, and continually improving the twin through data-driven insights. Just as importantly, they’ll need to work closely with domain experts (grid engineers, operators) to validate models and embed them in decision workflows – making sure that the AI recommendations are practical and trustworthy.

Ultimately, the **digital twin vision for OUC** is one of a smarter municipal utility: one that can anticipate problems before they occur, optimize performance in real time, and plan investments with digital foresight. By bridging previously siloed IT/OT systems and leveraging AI across generation, T&D, DERs, and customer services, OUC can enhance

resiliency and efficiency in serving the Orlando community. The journey will be iterative and collaborative, but the path is clear – as digital twin technology matures, it is poised to become an indispensable digital backbone for utilities aiming to innovate and adapt in the face of evolving energy challenges[7][46].

Sources: The insights and roadmap outlined above synthesize information from recent literature, industry case studies, and expert analyses on digital twin technology in the energy sector. Notable references include industry articles[45][26], technical guides[9], and utility pilot reports, which collectively underscore the technical foundations and strategic approaches for successful digital twin deployment in electric utilities. The specific citations (in brackets) correspond to these sources and provide additional details for further reading and validation of the points discussed.

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